

Descriptive Data Analytics Report: Zomato Restaurant Analysis :-

Analysing Restaurant Types, Customer Preferences, Rating and Online/Offline Ordering Behaviour

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Note on Data:

This report uses the publicly available Zomato Restaurant Dataset from India consisting of **148 restaurant entries and 7 business related features, and 12 months of data**. The dataset captures restaurant details, customer rating, voting patterns, online ordering, table booking availability, average cost for two people, and restaurant categories. These records are analysed to understand customer behavior, restaurant popularity, and ordering trends.

In today's digital world, online food delivery platforms have changed the way people order food. Among these platforms, **Zomato is one of the most popular restaurant discovery and food ordering applications in India**. It allows customers to explore restaurants, compare ratings, read reviews, place online orders, and make informed dining decisions. Every customer interaction on the platform generates valuable data that can be analyzed to understand customer behavior and restaurant performance.

This project focuses on **Descriptive Data Analytics** using a publicly available **Zomato Restaurant Dataset**. The **dataset contains information about restaurants, including restaurant type, customer ratings, number of votes, online ordering availability, and the approximate cost for two people**. By analyzing these features, meaningful insights can be obtained regarding customer preferences and restaurant performance.

The main purpose of this project is to understand customer ordering patterns and identify the factors that influence restaurant popularity. Different data visualization techniques have been used to present the findings in a simple and meaningful way. Graphs and charts make it easier to identify trends, compare different restaurant categories, and understand customer behavior.

The analysis begins by identifying **which type of restaurant is preferred by the majority of customers**. Different restaurant categories such as Dining, Cafes, Buffet, and Others are compared to understand customer

preferences. This helps identify which restaurant types are more popular on the platform.

The project also analyzes **customer voting patterns**. Votes indicate how many customers have rated a restaurant, making them an important indicator of a restaurant's popularity and customer engagement. Restaurants with higher votes generally have greater visibility and stronger customer trust.

Another important part of this project is the **rating distribution analysis**. Customer ratings help measure the quality of food and service provided by restaurants. By studying the rating distribution, it becomes possible to understand whether most restaurants provide satisfactory service or whether improvements are needed.

The dataset is also used to analyze the **average spending for two people**. This provides an understanding of customer spending behavior and helps identify the price range that is most common among restaurants. Such information can help businesses understand customer affordability and pricing strategies.

Another objective of this project is to compare **online and offline ordering behavior**. The **analysis examines whether restaurants offering online ordering receive better ratings than those without online ordering**. It also identifies which restaurant type receives the **highest number of offline orders**. These insights help understand customer preferences for ordering methods and support better business decisions.

The project demonstrates how **Python** can be used for data analysis. Libraries such as **Pandas**, **NumPy**, and **Matplotlib** are used for data cleaning, processing, analysis, and visualization. These tools make it easier to convert raw data into meaningful information that supports decision-making.

Overall, this **report provides a clear understanding of restaurant performance, customer preferences, voting behavior, rating trends, pricing patterns, and online/offline ordering behavior**.

1.Objective

The objective of this report is to analyze the Zomato dataset and answer the following business questions:

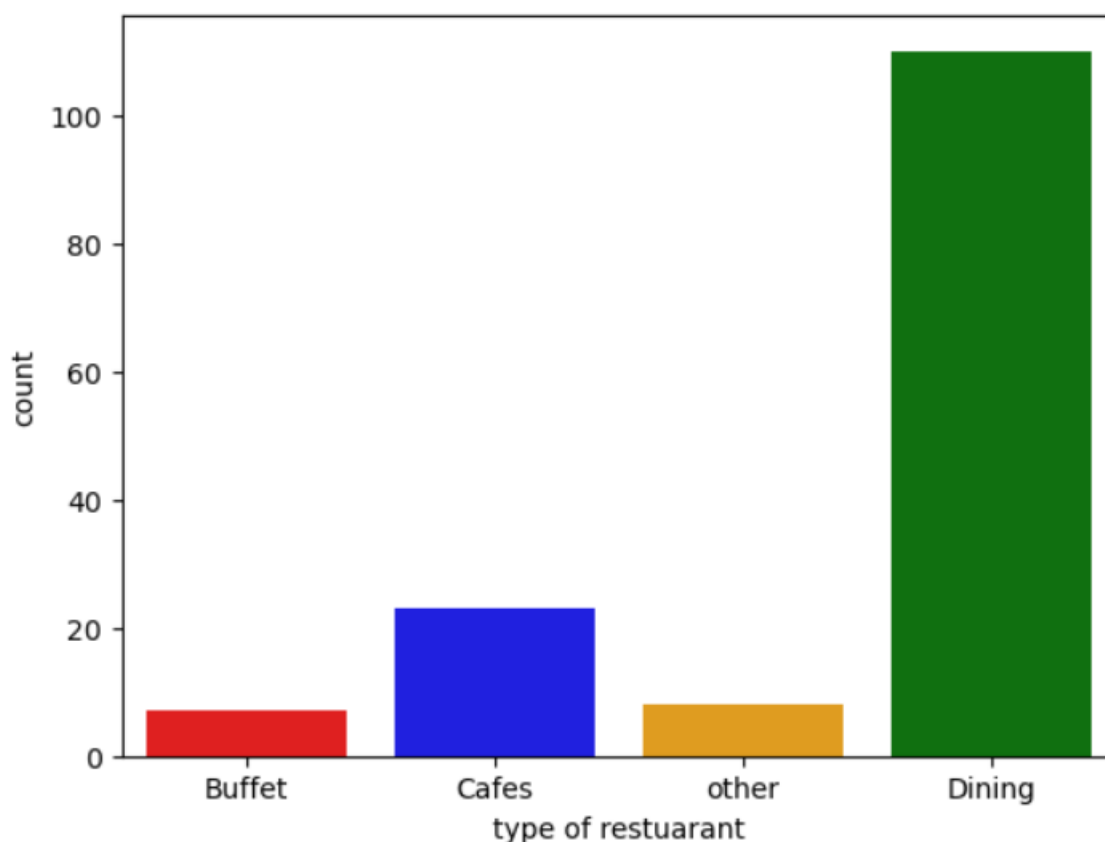
- 1) What type of restaurant do the majority of customers order from?
- 2) What are the ratings that the majority of restaurants have received?
- 3) Zomato has observed that most couples order online and their average spending on each other?
- 4) Which mode online and offline has received the maximum rating?
- 5) Which restaurant type is most preferred for offline orders?

2. How This Would Work At Real Scale

Zomato collects and analyses daily order data to track sales, customer behaviour, restaurant performance, and delivery trends, enabling better business decision through real – time analytics.

Layer	Purpose	Typical tools
Investigation	Customer Order analysis	Google
Stream Processing	Process customer orders and restaurant update	Python
Storage	Store order details, rating and cost information for future analysis	CSV File/Excel
Batch Aggregation	Restaurant data analysis, online and offline order mode	Python(pandas)
Serving/BI Layer	Visualise insight using charts and Report	Matplotlib,Seaborn,Numpy

3. Types of Restaurant



The graph shows that **Dining** is the most common restaurant type in the most common restaurant type in the dataset , with a count of around **102 restaurant**.

Cafes are the second largest category with around **20 restaurants**, while **Buffet (around 7)** and **Other (around 4)** have the lowest counts.

The distribution is highly concentrated in the Dinning category, indicating that most restaurants listed on Zomato belonging to this type in comparison,

Cafes have a moderate presence, whereas **Buffet** and Order restaurant types represent only a small portion of the dataset.

The **Restaurant Type** graph represents the distribution of different restaurant categories available in the Zomato dataset. The main objective of this analysis is to identify which type of restaurant is more common and preferred by customers. Different restaurant categories such as **Dining, Cafes, Buffet, and Others** have been compared to understand their presence in the dataset.

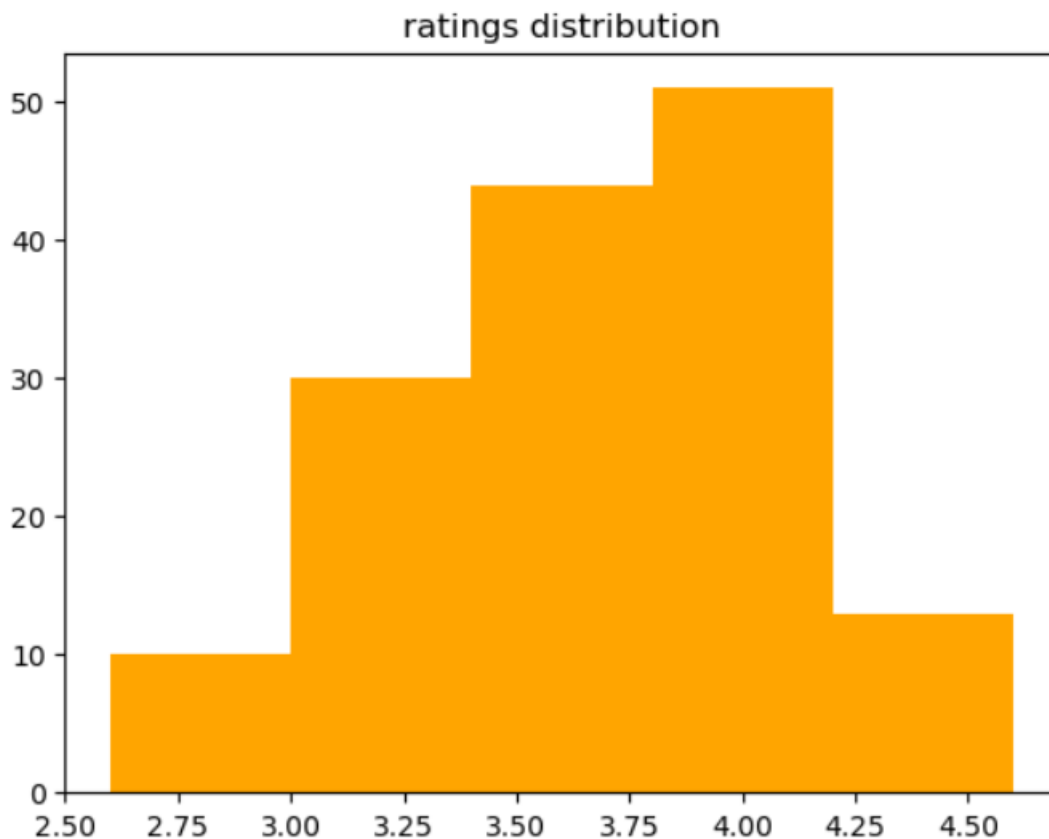
From the graph, it is clearly observed that **Dining restaurants** have the highest count among all restaurant types. This indicates that customers mostly prefer restaurants that provide complete meals and a comfortable dining experience. **Cafes** have the second-highest count, showing that they are also popular among customers for coffee, beverages, snacks, and casual meetings. On the other hand, **Buffet** and **Other** restaurant categories have comparatively fewer restaurants, indicating lower customer demand compared to Dining and Cafes.

This analysis helps understand customer preferences and restaurant distribution on the Zomato platform. The findings suggest that Dining restaurants play a major role in attracting customers, while Cafes also maintain a significant presence. These insights can help restaurant owners improve their services, plan business strategies, and help Zomato recommend restaurants more effectively based on customer demand.

KPI Definition Used:

- **Restaurant Count** = Total number of restaurant category.
- **Restaurant Distribution** = Number of restaurant grouped by type (Dinning , Cafes, Bufffet and Other).

4. Rating Of Restaurant



The rating distribution shows that most restaurants have **ratings between 3.5 and 4.2**, indicating overall good customer satisfaction.

The **highest concentration of ratings is around 4.0**, while only a few restaurants have **ratings below 3.0 or above 4.5**.

This suggests that the majority of restaurants provide consistent service and quality.

The **Restaurant Rating Analysis** helps evaluate the quality and performance of restaurants based on customer feedback. Customer ratings are one of the most important indicators of restaurant performance because they reflect customer satisfaction with food quality, service, cleanliness, and overall dining experience. In this project, the distribution of restaurant

ratings has been analyzed to understand how customers rate different restaurants listed on the Zomato platform.

From the graph, it is observed that **most restaurant ratings fall between 3.5 and 4.2**, indicating that the majority of restaurants provide good service and maintain satisfactory food quality. The **highest concentration of ratings is around 4.0**, which suggests that customers are generally satisfied with their dining experience. Restaurants with ratings in this range are likely to gain customer trust and attract repeat customers.

The graph also shows that only a **small number of restaurants have ratings below 3.0**. This indicates that very few restaurants receive poor customer feedback. Such restaurants may need to improve their food quality, customer service, hygiene, and overall dining experience to increase customer satisfaction. Similarly, **only a limited number of restaurants have ratings above 4.5**, showing that excellent ratings are achieved by only a few restaurants that consistently provide outstanding service.

The rating distribution appears balanced, without extreme variations, which suggests that most restaurants maintain a consistent standard of quality.

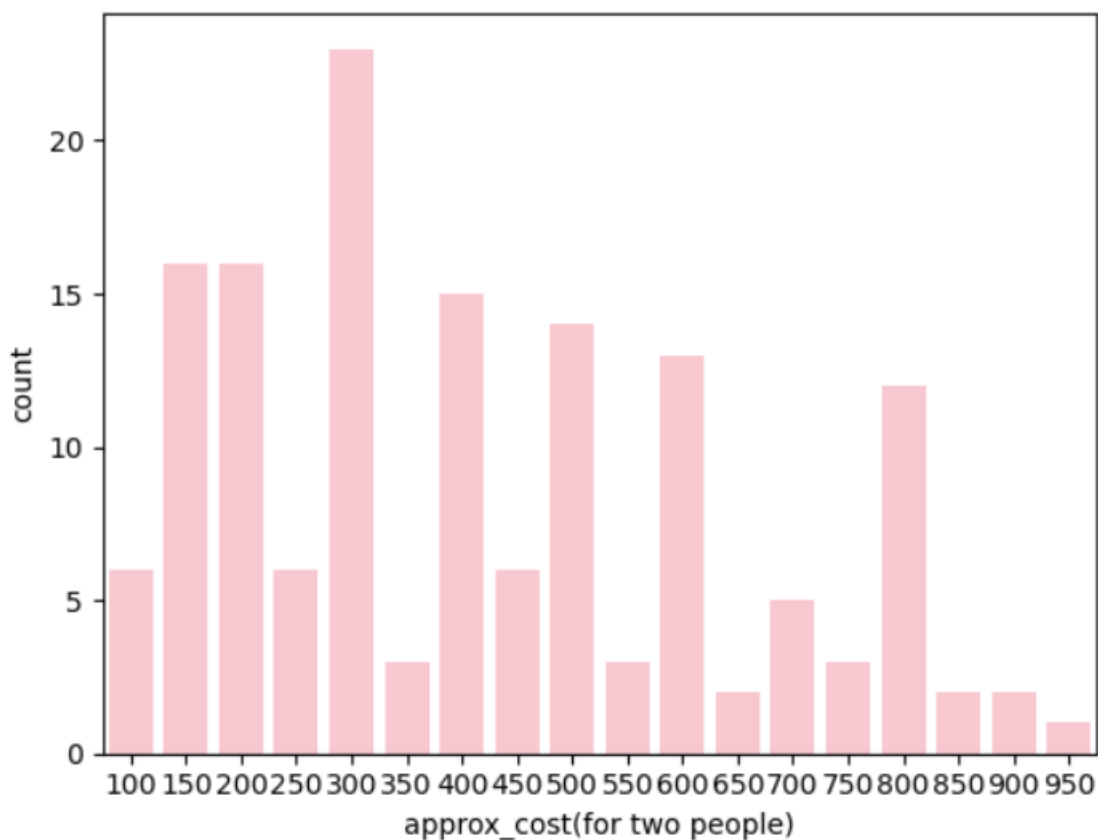
Customers generally prefer restaurants with higher ratings because ratings help them make informed decisions before placing an order or visiting a restaurant. **Therefore, maintaining a good rating is important for improving restaurant visibility and customer confidence.**

This analysis is valuable for both restaurant owners and the **Zomato platform**. Restaurant owners can use customer ratings to identify areas that require improvement and enhance the overall customer experience. At the same time, Zomato can recommend highly rated restaurants to customers,

improving customer satisfaction and encouraging repeat usage of the platform.

Overall, the **Restaurant Rating Analysis** provides meaningful insights into customer satisfaction and restaurant performance. **It helps identify rating trends, understand customer preferences, and supports better business decisions.** By continuously monitoring customer ratings, restaurants can improve their services, build customer loyalty, and maintain a strong reputation in the competitive food delivery market.

5. Average Order Spending By Couples



The graph shows the distribution of the approximate cost for two people.

Most couples spend between ₹ 250 and ₹350 , which is the most common spending range. A **moderate number of customer spend ₹400 – ₹600** , while **only a few orders are above ₹800**. This indicate that customers generally prefer restaurants that offer good food at affordable prices.

- Identifying the most preferred spending range among couples.
- Understanding customer spending behavior.
- Planning discounts and combo offers for popular price ranges.
- Supporting restaurant pricing and promotional strategies.

The spending pattern also indicates that most customers prefer restaurants that provide good value for money rather than choosing very expensive options. **The higher frequency of orders in the ₹250–₹350** price range suggests that this price segment is the most attractive for couples. As the **cost increases beyond ₹600**, the number of customers gradually decreases, showing that fewer people choose high-priced restaurants for regular dining.

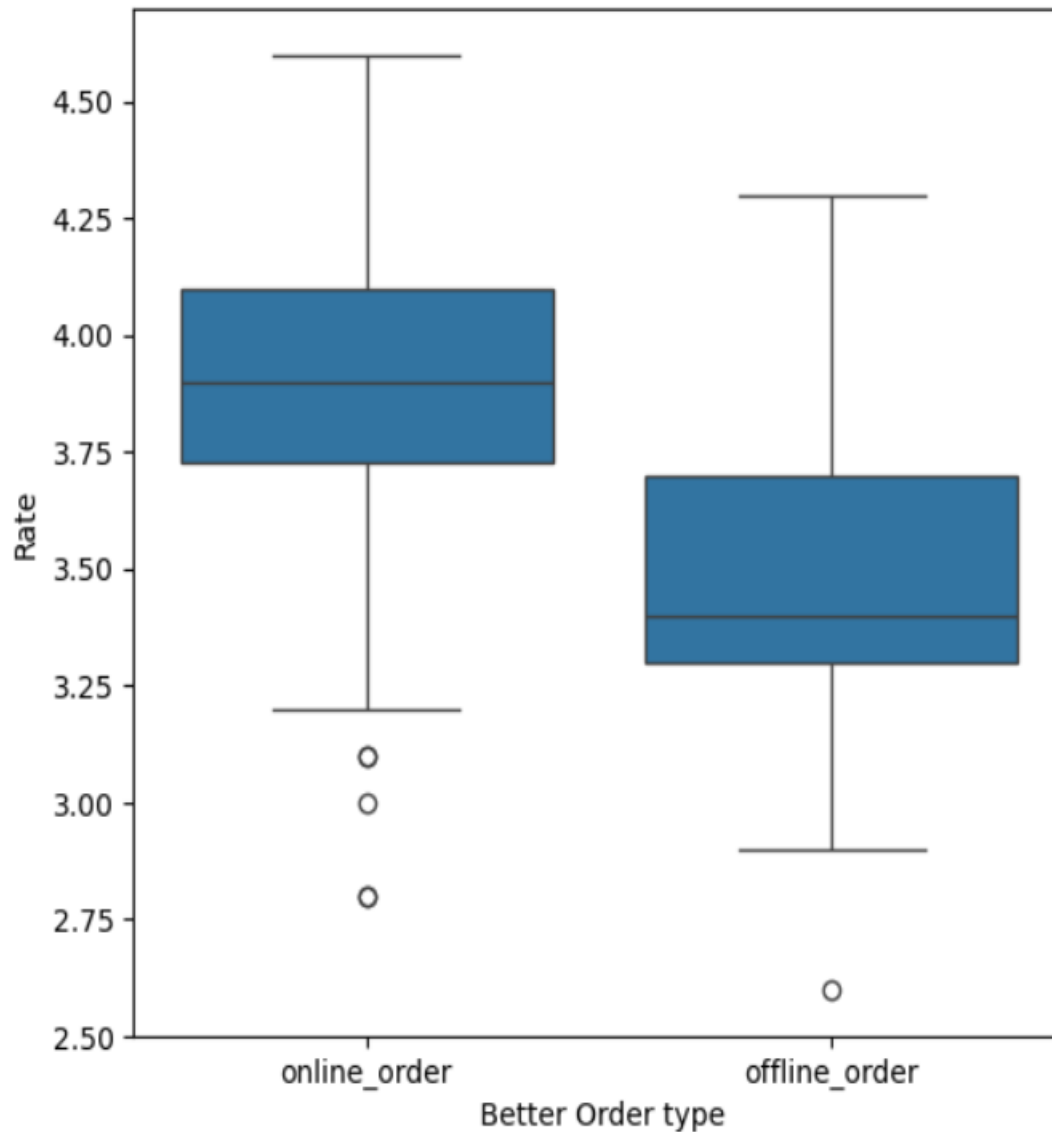
This analysis helps restaurants understand customer budget preferences and plan their pricing strategies accordingly. Restaurants can introduce combo meals, seasonal offers, or discounts in the most popular price range to attract more customers. They can also improve premium dining experiences to encourage customers to spend more on special occasions.

The **Average Order Spending by Couples** analysis helps understand how much couples usually spend while dining at restaurants listed on the **Zomato platform**. This analysis provides insights into customer spending behavior and identifies the most preferred price range. Understanding spending patterns helps restaurants design better pricing strategies and promotional offers.

From the graph, it is observed that **most couples spend between ₹250 and ₹350 per order**, making this the most common spending range. A moderate number of customers spend between **₹400 and ₹600**, while only a few customers spend **more than ₹800**. This indicates that customers generally prefer restaurants that offer good quality food at affordable or mid-range prices rather than expensive dining options.

The analysis shows that budget-friendly and mid-range restaurants attract a larger number of customers. Restaurants can use this information to introduce attractive combo meals, discounts, and special offers within the popular price range to increase customer satisfaction and sales. Overall, this analysis provides valuable insights into customer spending habits and supports better pricing, marketing, and business decisions.

6. Which mode maximum rating



The boxplot compares restaurant **rating based on Online and Offline order** modes. The online order mode (**Yes**) has a **higher median rating of around 3.9 – 4.0** while the offline order mode (**No**) has no **lower median rating of around 3.3 – 3.4** . Online orders also achieve the **highest rating, reaching approximately 4.6**, whereas **offline orders reach about 4.2**.

- Comparing customer satisfaction across online and offline orders.
- Identifying the order mode with better customer.
- Improving service quality for lower-rated order modes.
- Supporting decision for online promotions and customer experience.

The boxplot clearly shows that restaurants offering **online ordering** generally receive **better customer ratings** than restaurants that do not provide online ordering. The median rating of online orders is higher, which indicates that customers are more satisfied with the online ordering experience.

The spread of ratings for online orders is also more consistent, showing that most restaurants maintain good service quality. Although a few restaurants have lower ratings (outliers), the majority of online-order restaurants are rated between **3.7 and 4.1**, which represents good customer satisfaction.

In comparison, restaurants without online ordering have a lower median rating and slightly lower maximum rating. This may indicate that customers prefer the convenience of online ordering, including easy ordering, digital payment, faster service, and home delivery.

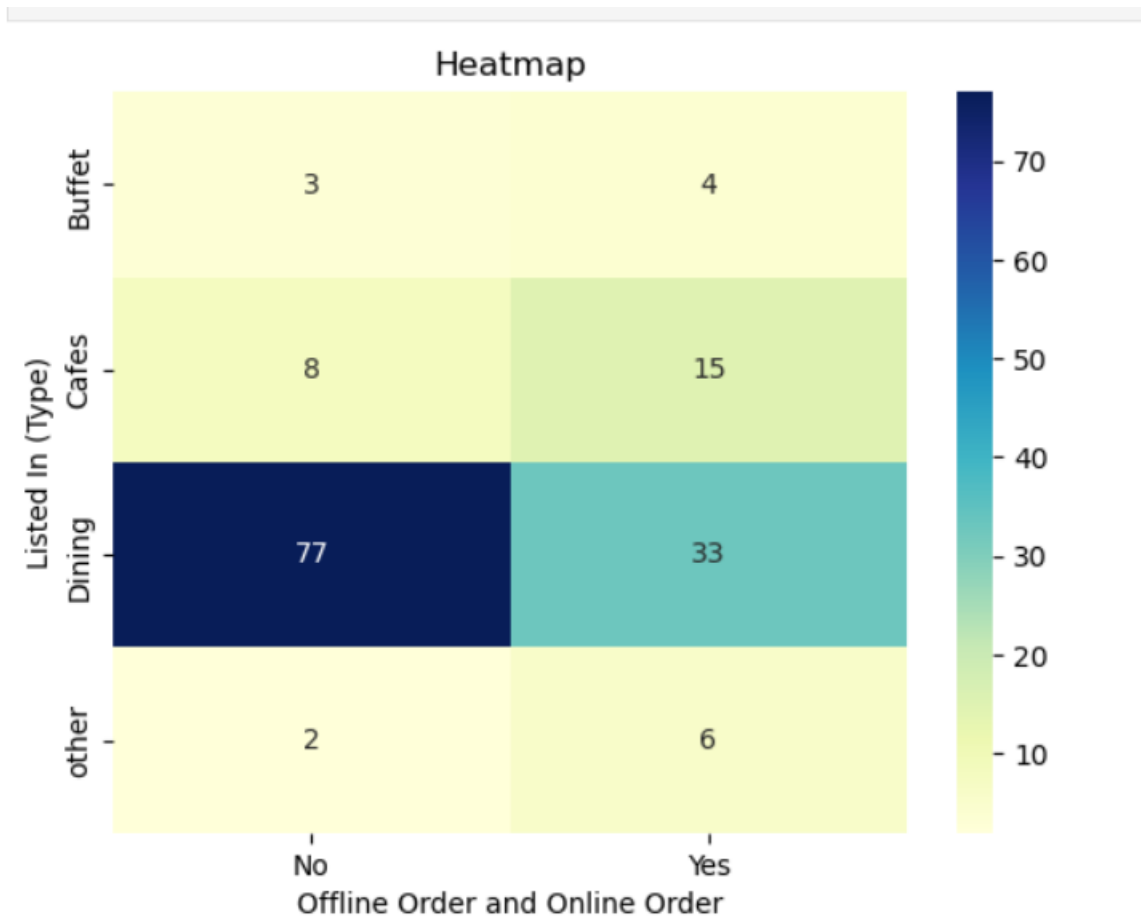
The **comparison between online and offline ordering** modes provides a better understanding of customer preferences and restaurant performance. The graph shows that customers are comfortable using both ordering methods, but the overall ratings indicate that restaurants offering **online ordering** tend to receive better customer feedback. This may be because online ordering provides

greater convenience, faster access to restaurant information, and easier order placement.

The results also suggest that customers consider factors such as delivery experience, food quality, order accuracy, and service while giving ratings. Restaurants that maintain consistency in these areas are more likely to receive higher ratings and positive customer reviews. On the other hand, restaurants with lower ratings should focus on improving service quality and customer satisfaction to build trust.

This analysis highlights the importance of maintaining high service standards regardless of the ordering mode. Restaurant owners can use these insights to improve both online and offline services, while **Zomato can recommend highly rated restaurants to customers based on their preferences.** Overall, the comparison helps businesses understand customer expectations and supports better operational and marketing decisions.

7. Restaurant type with highest Offline Orders



The heatmap compares restaurant types based on online ("**Yes**") and offline ("**No**") orders. It shows that **Dining restaurants have the highest number of offline orders (77)**, which is much higher than Cafes (8), Buffet (3), and Other restaurant types (2). This indicates that customers prefer visiting Dining restaurants in person rather than placing online orders.

Although Dining restaurants also receive **online orders (33)**, offline orders are significantly higher. Cafes show a moderate preference for online orders **(15) compared to offline orders (8)**, while Buffet and Other restaurant types have relatively low order counts in both categories.

The heatmap clearly highlights the difference between restaurant types and customer ordering preferences. The **Dining** category has the **highest number of orders, especially offline**, showing that many customers still prefer visiting dining restaurants for a better dining experience. This may be because **customers enjoy the restaurant atmosphere, service quality, and spending time with family or friends.**

On the other hand, **Cafes** receive more **online orders than offline orders.** **This suggests that many customers prefer ordering snacks, beverages, or light meals online** because it is more convenient and time-saving. **Buffet and Other restaurant categories have relatively fewer orders, indicating lower customer demand in this dataset.**

Overall, the analysis shows that customer ordering behavior changes depending on the type of restaurant. Restaurants can use this information to improve their marketing strategies, increase customer engagement, and provide better services according to customer preferences.

The heatmap provides a clear visual comparison of restaurant types based on **online ("Yes")** and **offline ("No")** ordering. It helps identify customer ordering preferences by displaying the number of restaurants in each category. Instead of analyzing individual values separately, the heatmap makes it easier to compare different restaurant types and understand overall ordering patterns.

From the heatmap, it can be observed that **Dining restaurants have the highest number of offline orders**, which indicates that customers still prefer visiting these restaurants in person. Dining restaurants usually offer a comfortable atmosphere, a wider menu, and a better dining experience,

making them a popular choice for families, friends, and group gatherings. This explains why the offline order count is much higher than other restaurant categories.

The heatmap also shows that **Cafes receive a comparatively higher number of online orders than offline orders**. This suggests that many customers prefer ordering snacks, beverages, and light meals online because it is more convenient and saves time. In contrast, **Buffet** and **Other** restaurant types have lower values in both online and offline ordering, indicating that they serve a smaller customer base within the dataset.

Another important observation is that customer ordering behavior is not the same across all restaurant types. Different restaurant categories attract customers for different reasons. While **Dining restaurants** are mainly chosen for **dine-in experiences**, **Cafes** often preferred for **quick orders and takeaway services**. This variation helps businesses understand which ordering mode is more suitable for each restaurant category.

The findings from the heatmap can help restaurant owners improve their business strategies. Restaurants with strong offline demand can focus on improving seating capacity, customer service, cleanliness, and the overall dining experience. At the same time, restaurants with higher online demand can invest in faster delivery services, attractive online offers, and better packaging to improve customer satisfaction.

8. Summary Of Key Metrics

Matric	Observation
Most common Restaurant Type	Dining
Average Restaurant Rating	Approximately 3.7
Most Common Rating Range	3.5 – 4.2
Average Cost for Two	Mostly ₹300 - ₹600
Better Rated Order Mode	Online Order = Yes
Highest offline order	Dinning

- Dining restaurant received the highest number of customer votes.
- Most **restaurant ratings are between 3.5 and 4.2**, indicating generally good customer satisfaction.
- Customer mostly spend ₹300 - ₹600 for two people.
- Restaurants offering online ordering have slightly higher rating than those without online ordering.

9. Next Steps to Move From Descriptive – Predictive

This report explains the current restaurant trends, customer rating, votes, and spending patterns. The next step is to use predictive analytics to forecast future outcomes.

- **Restaurant Popularity Prediction** – Predict which restaurant type(Dining, Cafes, Buffet etc.) is like to receive more customer in the future.
- **Customer Rating Prediction** - Predict future restaurant ratings based on customer reviews and past ratings.
- **Order Spending Forecast** – Estimate the average spending of customer for two people in upcoming periods.
- **Customer demand Forecasting** – Predict future customer demand to help restaurants manage seating and inventory.
- **Recommendation System** – Suggest the best restaurant to customer based on ratings, votes, and spending preferences.

10. References

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4. **Numpy Library** – Numerical Computation
5. **Matplotlib Library** – Data Visualization
6. **Seaborn Library** – Statistical Data Visualization
7. **Jupyter Notebook** – Data Analysis
8. **Microsoft Word** – Report Documentation